

AWI-ESM3 ↔ PISM orography and SMB coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-08-03

Abstract

Mission. Close the atmosphere half of the AWI-ESM3 ↔ PISM cycle in both directions: PISM’s surface mass balance driven by the OIFS leg that just ran, and the ice-sheet surface elevation OIFS runs with following the ice sheet PISM produced. The ocean half was closed by the moving-cavity campaign; this is its companion.

Where we are. The ISM-mapper builds and links as an `esm_tools` component, the PISM grid and all three coupling links are generated and verified, and the OASIS exchange is declared, but nothing has yet run coupled.

Timeline in §1, current state and open items in §2, dead claims in §3.2.

1 The campaign, in the order it happened

1.1 Taking stock: the atmosphere side is a stub (2026-08-03)

`couplings/oifs/just_copy_stuff.functions` copies `atmosphere_file_for_ice.nc` verbatim out of `a270122/lars_an`. Confirmed in `eval5`: a 192×96 ECHAM Gaussian grid carrying `temp2`, `tsurf`, `aprt`, `orog`, `swd`, `cc`, `q2m`, `tau`, `emiss`, `uv`, `TOAswd`.

TAINTED: every PISM leg of the moving-cavity campaign, `eval5` and `fgdbg02` included, ran dEBM on a 130 ka paleo ECHAM atmosphere. No PISM surface mass balance number from that campaign is a statement about AWI-ESM3. Ocean-side results are not affected; they are forced from FESOM output directly.

Two further findings from the same read. `suorog.F90:151-250` already implements both returns: `ECE_CPL_ISMM` reads `usurf_oifs` and patches `ZZOROG` through `REESPE/SPEREE`, `ECE_LANDICE` reads `plit_oifs`. That code is upstream (vendor `a2e46d5`, Jorge Bernales) and our `suorog.F90` is byte-identical to it. But we run `ECE_LANDICE` only, and the `plit` we write is not PISM’s ice mask: `make_plit_oifs.py` derives it from the regenerated ICMGG land–sea mask capped at 60°S . `eval5/couple/antar_pism2ece.nc` holds one variable, `plit_oifs`, `nx=40320`. No `usurf_oifs`. So no PISM geometry reaches the atmosphere at all.

1.2 Choosing the ISM-mapper over XIOS (2026-08-03)

Everything dEBM needs is already in the XIOS output, and the de-accumulation pattern exists in `field_def_ljpg_safe.xml.j2`. Checked against `eval5` year 1903: `ssrd` = 6.845×10^5 and `tisr` = $1.226 \times 10^6 \text{ J m}^{-2}$ per one-hour step give 190 and 340 W m^{-2} ; `lsp+cp` = $1.13 \times 10^{-4} \text{ m/step}$ gives $3.14 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$. All correct global means, so a jinja-gated `file_def` block would have delivered the forward direction with no new component.

Decided against it. XIOS solves the forward direction only: it cannot deliver `plit` into the live atmosphere, and cannot route basal melt and discharge into the ocean. Both need a component inside the OASIS exchange, and one mechanism is better than two. Staying on the EC-Earth component also keeps CISSEMBEL available.

Also decided: carry both SMB schemes with dEBM as default; SMB before orography; everything behind `general.smb_coupled`.

1.3 The ISM-mapper becomes an `esm_tools` component (2026-08-03)

`configs/components/ismm/ismm.yaml`, in the `develop-is` coupling, always built.

It is not a standalone repository: it lives inside `ecearth4` under `sources/ism-mapper`, so the component clones `ecearth4` and builds that subtree. The clone is 37 MB, not the 663 MB the local `.git` suggests, because every `sources/*` model is a submodule that is not recursed into. The branch `cissembel-levante-intel` is required rather than preferred: it carries 5799a2a, which fixes an `EBMfinalize` use-after-free that crashes any run with a non-CISSEMBEL region.

CISSEMBEL is a *link-time* dependency even on the direct path (`ism-mapper_run_mod.F90:7` has an unguarded `use modEBMworld`), so it is built from the same clone. At runtime `EBMworld` only fires for regions with `ISMrgncis(n)=.true.`, so it never executes for us.

The committed `Makefile` carries ECMWF HPC2020 paths. Everything it needs is a plain make variable and command-line macros beat file assignments, so the machine configuration is passed in `comp_command` rather than by patching a tracked file.

Verified. Built end to end from a fresh clone. The open risk was whether code written against OASIS 5.2 links against our 5.0-smhi; it does, with no source change. The binary carries `oasis_init_comp`, `oasis_def_var`, `oasis_enddef`, `ebmworld` and `ebminitialize`.

1.4 The PISM grid goes into `ocp-tool` (2026-08-03)

The ISM-mapper is an OASIS component, so the PISM domain must appear in `grids.nc/masks.nc/areas.nc`. This went into `ocp-tool`, not `esm_tools`: `ocp-tool` already sits between PISM and AWI-ESM3, it is where the weight generation lives, and `ocp-tool/grids/` already has the abstraction (a class with `cell_latitudes`, `cell_longitudes`, `cell_corners`, `cell_areas`, `cell_masks`). Putting it there deleted a planned step: `ocp-tool` writes `ismp` itself, so nothing appends fragments afterwards.

Centres come from the PISM file's own `lat/lon`; corners are inverse-projected from the cell corner (x, y) ; areas are spherical excess over two triangles.

Verified on `antarct_cr` 8 km, 761×761 , three ways, because a wrong grid fails as silently-bad weights rather than a crash:

- Area total 36.03 against $36.28 \times 10^6 \text{ km}^2$ from the plane area corrected by the polar stereographic scale factor, -0.70% .
- Cell areas 67.0 km^2 at the pole to 53.8 at the outer corner against a nominal 64, the right sign and size for `lat_ts = -71`.
- All 579121 cells wind counterclockwise seen from outside the sphere, no sign flips. SCRIP requires this and the area check cannot detect it.

A full TCO95/CORE3 run puts `ismp` in all three files, bit-identical to an independently generated reference.

Found on the way: `ocp-tool` was appending the FESOM grid unconditionally. FESOM writes its own `feom` entry at runtime in domain-decomposition order; `ocp-tool`'s is mesh order, and the two disagree. Now behind `ocean.write_oasis_grid`, default off. *This changes default output for every FESOM config, not only ours.*

1.5 Declaring the OASIS coupling (2026-08-03)

Field names on the ice side come from the ism-mapper itself (cplng_config_mod.F90): <prefix>_Precip_liquid, _Precip_solid, _Evaporation, _Runoff, _Soil4T, _SST inbound and <prefix>_plit out. On the OIFS side A_Evap_ISM, A_SST_ISM and A_Soil4T_ISM are what ecearth.F90 declares for the direct path; precip and runoff are reused from the ocean path.

Two constraints the parser caught, both of which would have been expensive later:

- A_Runoff is on atmr, not atma, so it needs its own exchange line and its own direction. Grouping it with precip fails the same-grid validation.
- include_models is consumed before general-level choose_ blocks resolve, so ismm must sit in the develop-is add_include_models. It moves together with smb_coupled: a launched ismm with no coupled fields deadlocks OASIS.

1.6 Generating the weights (2026-08-03)

Weights are not left to OASIS at model runtime. ocp_tool/oasis_weights.py precomputes them with pyOASIS taken from the coupled model's own OASIS build, ported from rdy2cpl: one MPI rank per link, each calling enddef() so SCRIP writes that link's rmp_*.nc. Geometry is read back from the OASIS description files, so the ismp entry of §1.4 is exactly what a link consumes. It needs compute nodes; the one-rank-per-link parallelism is what makes it tractable.

Tested on 3 nodes against ocp-tool's own TCO95/CORE3 output:

link	map	result
A096 → ismp	BILINEAR D	69 MB
R096 → ismp	GAUSWGT D	122 MB
ismp → A096	GAUSWGT LR	44 MB
ismp → A096	DISTWGT LR	34 MB
ismp → A096	BILINEAR	OASIS abort, rc=1
ismp → A096	CONSERV	OASIS abort, rc=1

Why the ice→atm conservative and bilinear maps die. 381 of the 579121 cells have a corner longitude spread above 180° along the polar seam, and the pole cell's four corners sit at −135, 135, 45, −45, encircling the pole outright. That breaks SCRIP's per-cell longitude line integral (CONSERV) and its enclosing-quadrilateral search (BILINEAR). Neighbour-based maps do not care. Same geometry as the reason the areas use spherical excess.

Consequence. plit reaches suorog as an interpolated value, not the area fraction ECE_LANDICE_THRESH is written to read. EC-Earth's bilinear would give the same, so we are no worse than the reference, but the threshold semantics are looser than ideal.

2 State of the coupling (2026-08-03)

2.1 Proven

- ISM-mapper builds from a clean clone and links against our OASIS 5.0-smhi.
- ismp grid written by ocp-tool into all three OASIS files, geometry verified three ways (§1.4).
- All three coupling links generate on compute (§1.6).
- The esm_tools config assembles: ismm in process_ordering, rstism.nc a recognised OASIS restart, four exchange lines resolving with the prefix substituted.

2.2 Open

- **Nothing has run coupled.** No leg has been launched with `ismm` in it.
- **ECE_CPL_ISMM not yet set.** It belongs in the runscript next to `ECE_LANDICE/ISMRGNPFX`; `NAMECECFG` is add'd per chunk there and a second `add_` from the setup collides.
- **PISM links not wired into `regenerate_oasis_weights`,** which still defaults to the `feom` set. Also `nlinks` there defaults to 4 while `awiesm3_feom_links()` returns 5, so one link already serialises; PISM takes it to 8.
- **Per-leg `namelist.ismm` and the `cpl_dir/{aux,prev_year}` staging.**
- **Consume the SMB:** turn `{region}_smb_{year}.nc` into `atmosphere_file_for_ice.nc`, replacing the stub of §1.1.
- **orog for the downscaling reference.** `field_def.xml:176` has a `z` but it is in the `3D_ml` group, so model-level geopotential. Source it from the ICMGG with eccodes for param 129, as `make_plit_oifs.py` does for `lsm`.
- **usurf_oifs writer** and replacing the ICMGG-derived `plit` with PISM's own mask.
- **Freshwater return.** PISM basal melt and discharge do not reach FESOM. Not needed for SMB or orography, but it is the remaining leg of the mass budget.
- **atlas 0.31.1 → 0.39.0** came in with the vendor merge and is unbuilt.

3 Reference

3.1 Secondary fixes

what	where	commit
<code>mod_domain.F08</code> listed in <code>MOD_SRC0</code> but absent, CISSEMBEL unbuildable	<code>ecearth4</code>	<code>41e0d8e</code>
Runoff / runoff-mapper exchanges made optional (-huge trap)	<code>ecearth4</code>	<code>f9d0349</code>
<code>ECE_ISMM_SET_STATE</code> call had no <code>intfb</code> include	<code>oifs-48r1</code>	<code>38c3ab0</code>
<code>debm</code> missing from <code>add_other_components</code> , <code>esm_master</code> aborted	<code>esm_tools</code>	<code>55a7935</code>

The -huge trap is worth stating once in full: receive buffers are initialised to -huge, `AccumulateECEForcingFields` adds every `OASIS_IN` field with no check that it is coupled, and an uncoupled field is never written. So a declared-but-unpartnered input accumulates -huge for a whole leg. This is the same failure class as the OIFS-side -HUGE first-send bug, arriving from the other end.

3.2 Falsified claims (kept, struck through)

- ~~Adding a PISM grid means teaching the per-leg regeneration to emit SCRIP weights to and from TCO95 on every cycle. This is the expensive part and the main schedule risk. Overstated: the weight machinery is general, so PISM costs a few Link entries.~~
- ~~OASIS computes the weights itself at runtime, and the PISM grid does not move, so there is no per-cycle work.~~ Wrong twice. Weights are precomputed via `pyOASIS` (§1.6), and the links do regenerate per mesh-change leg because the OIFS mask moves where PISM is.
- ~~CONSERV is the right map for plit, since ECE_LANDICE_THRESH reads an area fraction. It aborts on a pole-centred grid (§1.6). GAUSWGT is what works.~~
- ~~The ice→atm link is BILINEAR-D.~~ The SCRIPR letter is the *source* grid type and `ismp` is logically rectangular, so LR; and `BILINEAR` fails here regardless.

3.3 Method notes

- `esm_runscripts -c` assembles the full runtime config and catches grid-consistency and component-loading errors. Both config bugs above were found this way, not by reading.
- `esm_master -check <target>` renders the build without executing it. It also revealed the pre-existing `debm` section-header failure.
- Weight generation must run from a shared filesystem. A first attempt with the working directory on `/tmp` died in 14s: `/tmp` is node-local on Levante, so the compute node could not write the job output.
- Grid winding is invisible to an area check, which takes an absolute value. Test it separately by dotting the corner-polygon normal with the outward radial vector.

3.4 Assets

what	where
Reference document	<code>esm_tools/ece4-pism-coupling.md</code>
PISM OASIS grid (reference copy)	<code>/work/ab0246/a270092/input/pism/oasis/</code>
Weight-generation test output	<code>/work/ab0246/a270092/tmp/ismg_wgen{,2}/</code>
ocp-tool ice-enabled config	<code>configs/TC095_CORE3_ICE.yaml</code>

repository	branch	state
<code>esm_tools</code>	<code>awiesm3-is-orography-and-smb-coupling</code>	<code>c94d11f89</code> local
<code>ocp-tool</code>	<code>feat/pism-oasis-grid</code>	PR #57, draft
<code>oifs-48r1</code>	<code>movcav-landice+co2-conc-driven</code>	pushed
<code>ecearth4</code>	<code>cissembel-levante-intel</code>	pushed